

APM1220: FA 2 - Foundations

Sample Geometry and Random Sampling

Problem 3.1

Given Data: The data matrix is given as $X = \begin{bmatrix} 9 & 1 \\ 5 & 3 \\ 1 & 2 \end{bmatrix}$. The matrix X is a 3×2 matrix, meaning there are $n = 3$ observations and $p = 2$ variables.

(a) Scatter Plot in $p = 2$ Dimensions and Sample Mean

To locate the sample mean, we average the columns of the data matrix X .

$$\bar{x} = \begin{bmatrix} \frac{9+5+1}{3} \\ \frac{1+3+2}{3} \end{bmatrix} = \begin{bmatrix} \frac{15}{3} \\ \frac{6}{3} \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

Final Answer (a): In a $p = 2$ dimensional Cartesian plane, the data points are plotted at $(9, 1)$, $(5, 3)$, and $(1, 2)$. The sample mean $\bar{x} = \begin{bmatrix} 5 & 2 \end{bmatrix}'$ acts as the center of gravity and is located at the coordinate $(5, 2)$.

(b) $n = 3$ -Dimensional Representation and Deviation Vectors

We extract the columns of X into $n = 3$ dimensional vectors: $y_1 = \begin{bmatrix} 9 & 5 & 1 \end{bmatrix}'$ and $y_2 = \begin{bmatrix} 1 & 3 & 2 \end{bmatrix}'$.

The mean vectors in 3D space are calculated by multiplying the scalar mean by a 3×1 vector of ones, $\mathbf{1}$:

$$\bar{x}_1 \mathbf{1} = 5 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}, \quad \bar{x}_2 \mathbf{1} = 2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix}$$

Subtracting these (dimensionally conformant 3×1 vectors) yields the deviation vectors $d_i = y_i - \bar{x}_i \mathbf{1}$:

$$d_1 = \begin{bmatrix} 9 \\ 5 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}, \quad d_2 = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

Final Answer (b): The deviation vectors are $d_1 = \begin{bmatrix} 4 & 0 & -4 \end{bmatrix}'$ and $d_2 = \begin{bmatrix} -1 & 1 & 0 \end{bmatrix}'$. In an $n = 3$ sketch, these vectors emanate from the tips of the mean vectors $\bar{x}_1 \mathbf{1}$ and $\bar{x}_2 \mathbf{1}$ pointing towards y_1 and y_2 .

(c) Lengths, Angle, and Relation to S_n and R

The squared lengths of the vectors are given by the inner product $d'_i d_i$ (multiplying a 1×3 row vector by a 3×1 column vector yields a scalar):

$$L_{d_1}^2 = d'_1 d_1 = (4)^2 + (0)^2 + (-4)^2 = 32 \implies L_{d_1} = 4\sqrt{2}$$

$$L_{d_2}^2 = d'_2 d_2 = (-1)^2 + (1)^2 + (0)^2 = 2 \implies L_{d_2} = \sqrt{2}$$

The inner product (dot product) of d_1 and d_2 is:

$$d'_1 d_2 = (4)(-1) + (0)(1) + (-4)(0) = -4$$

The cosine of the angle θ is:

$$\cos(\theta) = \frac{d'_1 d_2}{L_{d_1} L_{d_2}} = \frac{-4}{(4\sqrt{2})(\sqrt{2})} = \frac{-4}{8} = -\frac{1}{2}$$

Final Answer (c): The lengths are $L_{d_1} = 4\sqrt{2}$ and $L_{d_2} = \sqrt{2}$, with $\cos(\theta) = -\frac{1}{2}$. Geometrically, the squared lengths relate to the biased sample variance by $L_{d_i}^2 = nS_{n,ii}$. Thus, $32 = 3S_{n,11} \implies S_{n,11} = \frac{32}{3}$, and $2 = 3S_{n,22} \implies S_{n,22} = \frac{2}{3}$. The inner product relates to covariance: $-4 = 3S_{n,12} \implies S_{n,12} = -\frac{4}{3}$. The cosine of the angle is precisely the sample correlation coefficient: $r_{12} = -\frac{1}{2}$.

Problem 3.6

Given Data: $X = \begin{bmatrix} -1 & 3 & -2 \\ 2 & 4 & 2 \\ 5 & 2 & 3 \end{bmatrix}$ (Dimensions: 3×3).

(a) Matrix of Deviations and Rank

First, compute the mean vector (averaging the columns):

$$\bar{x} = \begin{bmatrix} 6 \\ 9 \\ 3 \\ 3 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

The matrix of means is calculated by multiplying a 3×1 ones vector by the 1×3 mean row vector:

$$\mathbf{1}\bar{x}' = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 1 \\ 2 & 3 & 1 \\ 2 & 3 & 1 \end{bmatrix}$$

Subtracting this from X yields the matrix of deviations:

$$X - \mathbf{1}\bar{x}' = \begin{bmatrix} -1-2 & 3-3 & -2-1 \\ 2-2 & 4-3 & 2-1 \\ 5-2 & 2-3 & 3-1 \end{bmatrix} = \begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix}$$

Final Answer (a): The deviation matrix is $X_c = \begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix}$. This matrix is **not** of full rank. The sum of the rows of any mean-corrected data matrix is always the zero vector, which means the rows are linearly dependent. Therefore, the rank is at most $n - 1 = 2$.

(b) Sample Covariance Matrix S and Generalized Variance

The sample covariance matrix is $S = \frac{1}{n-1} X_c' X_c$. Multiplying the 3×3 transpose by the 3×3 deviation matrix:

$$X_c' X_c = \begin{bmatrix} -3 & 0 & 3 \\ 0 & 1 & -1 \\ -3 & 1 & 2 \end{bmatrix} \begin{bmatrix} -3 & 0 & -3 \\ 0 & 1 & 1 \\ 3 & -1 & 2 \end{bmatrix} = \begin{bmatrix} 18 & -3 & 15 \\ -3 & 2 & -1 \\ 15 & -1 & 14 \end{bmatrix}$$

Dividing by $n - 1 = 2$:

$$S = \begin{bmatrix} 9 & -\frac{3}{2} & \frac{15}{2} \\ -\frac{3}{2} & 1 & -\frac{1}{2} \\ \frac{15}{2} & -\frac{1}{2} & 7 \end{bmatrix}$$

Final Answer (b): Because the deviation matrix is not of full rank, the covariance matrix S is singular. Thus, the generalized sample variance (the determinant) is $|S| = 0$. Geometrically, this means the sample deviation vectors span a 2-dimensional plane rather than a full 3-dimensional volume, resulting in a hypervolume of zero.

(c) Total Sample Variance

The total sample variance is the trace of matrix S . **Final Answer (c):** $\text{tr}(S) = s_{11} + s_{22} + s_{33} = 9 + 1 + 7 = 17$.

Problem 3.9

Corrected Data Matrix: $X = \begin{bmatrix} 12 & 17 & 29 \\ 18 & 20 & 38 \\ 14 & 16 & 30 \\ 20 & 18 & 38 \\ 16 & 19 & 35 \end{bmatrix}$

(a) Mean Corrected Matrix and Linear Dependence

The mean vector is $\bar{x} = [16 \ 18 \ 34]'$. Subtracting the mean matrix yields the mean corrected matrix X_c :

$$X_c = \begin{bmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{bmatrix}$$

Final Answer (a): Because column 3 is strictly the sum of column 1 and column 2, the columns are linearly dependent. The vector establishing this dependence is $a = [1 \ 1 \ -1]'$, such that $X_c a = \mathbf{0}$.

(b) Covariance Matrix S and Rescaling

We calculate $S = \frac{1}{n-1} X_c' X_c = \frac{1}{4} X_c' X_c$:

$$X_c' X_c = \begin{bmatrix} -4 & 2 & -2 & 4 & 0 \\ -1 & 2 & -2 & 0 & 1 \\ -5 & 4 & -4 & 4 & 1 \end{bmatrix} \begin{bmatrix} -4 & -1 & -5 \\ 2 & 2 & 4 \\ -2 & -2 & -4 \\ 4 & 0 & 4 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 40 & 12 & 52 \\ 12 & 10 & 22 \\ 52 & 22 & 74 \end{bmatrix}$$

$$S = \begin{bmatrix} 10 & 3 & 13 \\ 3 & \frac{5}{2} & \frac{11}{2} \\ 13 & \frac{11}{2} & \frac{37}{2} \end{bmatrix}$$

Final Answer (b): The generalized variance $|S| = 0$ because the matrix S is singular (rank 2). Testing the vector a :

$$Sa = \begin{bmatrix} 10 & 3 & 13 \\ 3 & \frac{5}{2} & \frac{11}{2} \\ 13 & \frac{11}{2} & \frac{37}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 10 + 3 - 13 \\ 3 + \frac{5}{2} - \frac{11}{2} \\ 13 + \frac{11}{2} - \frac{37}{2} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Since $Sa = 0a$, a acts as an eigenvector corresponding to the eigenvalue zero.

(c) Verify Linear Dependence

Final Answer (c): For every row j in the corrected data matrix, the relationship $1(x_{j1}) + 1(x_{j2}) - 1(x_{j3}) = 0$ holds perfectly. Thus, linear dependence exists with $a_1 = 1$, $a_2 = 1$, and $a_3 = -1$.

Problem 3.18

Given Data: Mean vector $\bar{x} = [0.766 \ 0.508 \ 0.438 \ 0.161]'$ and covariance matrix S .

(a) Total Energy Consumption

Define the transformation vector $a = \mathbf{1} = [1 \ 1 \ 1 \ 1]'$. Total energy mean:

$$\bar{y} = a' \bar{x} = 0.766 + 0.508 + 0.438 + 0.161 = 1.873 = \frac{1873}{1000}$$

The total variance is the sum of all elements in the 4×4 matrix S :

$$s_y^2 = \mathbf{1}'S\mathbf{1} = \sum_{i=1}^4 \sum_{k=1}^4 s_{ik} = 1.760 + 1.398 + 0.510 + 0.245 = 3.913 = \frac{3913}{1000}$$

Final Answer (a): The sample mean of total energy is 1.873 and the variance is 3.913.

(b) Excess Petroleum Over Natural Gas

Define the transformation vector $c = [1 \quad -1 \quad 0 \quad 0]'$. The mean of the excess ($v = x_1 - x_2$) is:

$$\bar{v} = c'\bar{x} = 0.766 - 0.508 = 0.258 = \frac{129}{500}$$

The sample variance is:

$$s_v^2 = c'Sc = s_{11} + s_{22} - 2s_{12} = 0.856 + 0.568 - 2(0.635) = 0.154 = \frac{77}{500}$$

The sample covariance with the total variable y is computed via $c'S\mathbf{1}$, which equals the sum of the first row of S minus the sum of the second row:

$$\text{Cov}(v, y) = \sum_{j=1}^4 s_{1j} - \sum_{j=1}^4 s_{2j} = 1.760 - 1.398 = 0.362 = \frac{181}{500}$$

Final Answer (b): The mean excess is 0.258, the variance is 0.154, and the covariance with total energy is 0.362.

Problem 3.20

Given Data: Dataset of $n = 25$ incidents.

(a) Summary Statistics Method

Calculating sums from the table: $\sum x_1 = 281.9$ and $\sum x_2 = 435.4$. The individual variable means are:

$$\bar{x}_1 = \frac{281.9}{25} = 11.276, \quad \bar{x}_2 = \frac{435.4}{25} = 17.416$$

The mean of the difference $d = x_2 - x_1$ is $\bar{d} = 17.416 - 11.276 = 6.14$.

To calculate the variance of the difference from summary statistics, we need s_{11} , s_{22} , and s_{12} :

$$\begin{aligned} \sum x_1^2 = 3732.53 &\implies s_{11} = \frac{3732.53 - 25(11.276)^2}{24} = \frac{553.8156}{24} \\ \sum x_2^2 = 9604.20 &\implies s_{22} = \frac{9604.20 - 25(17.416)^2}{24} = \frac{2021.2736}{24} \end{aligned}$$

$$\sum x_1 x_2 = 4861.90 \implies s_{12} = \frac{4861.90 - 25(11.276)(17.416)}{24} = \frac{-47.668}{24}$$

Using the linear combination variance formula $s_d^2 = s_{11} + s_{22} - 2s_{12}$:

$$s_d^2 = \frac{553.8156 + 2021.2736 - 2(-47.668)}{24} = \frac{2670.44}{24} = \frac{66761}{600} \approx 111.268$$

Final Answer (a): Mean difference is exactly $\frac{307}{50} = 6.14$, and the variance is exactly $\frac{66761}{600} \approx 111.268$.

(b) Individual Values Method

We transform the bivariate data into a univariate dataset $d_j = x_{j2} - x_{j1}$ for all $j = 1, \dots, 25$. Summing the individual differences gives $\sum d_j = 153.5$.

$$\text{Mean } (\bar{d}) = \frac{153.5}{25} = 6.14$$

Summing the squared differences gives $\sum d_j^2 = 3612.93$.

$$\text{Variance } (s_d^2) = \frac{\sum d_j^2 - 25(\bar{d})^2}{24} = \frac{3612.93 - 25(6.14)^2}{24} = \frac{2670.44}{24} = \frac{66761}{600}$$

Final Answer (b): Both the mean and variance computed directly from the individual values match the summary statistics methodology exactly. Mean = 6.14 and Variance = $\frac{66761}{600}$.